centroid to each pair. If you read the book thus far, by now you should feel goose bumps when you hear “brute force” and, as a conditional reflex, think: can we do any better? We’ll talk again about this question a couple of sections down the road.

For the moment, let’s see the latest helper method we need to implement to complete k-means algorithm, the one updating centroids, described in listing 12.4.

Listing 12.4 updateCentroids

```
function updateCentroids(points, clusterIndices)
                  #1
  centroids ←
[]
#2
  for cIndex in uniqueValues(clusterIndices)
do                                  #3
    for j in {0, .., dim-1}
do                                                  #4
      centroids[cIndex][j] ←
                mean({points[k][j] | clusterIndices[k] ==
cIndex})    #5
  return centroids
                                                                #6
```

#1 Method `updateCentroids` takes the list of points in the dataset and the classification of points WRT current clusters (centroids). It returns the list of the centroids computed for each cluster.
#2 Initialize the array of centroids.
#3 Goes through all (unique) centroids’ indices. Assuming clusters’ indices goes from `0` to a certain value `m`, without